

Resistor Value Calculation Standalone Application with MATLAB

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We know that MATLAB is a very useful software tool and with its help we can build different types of applications. But what if we want to distribute the code or application to people who do not have MATLAB installed in their systems, or if we want to run a large number of MATLAB jobs simultaneously on a computing cluster and we only have a limited number of software with valid licence? In such situations, what we need is a standalone application.

Here we describe the procedure to make a MATLAB standalone application with a resistor value calculator utility program. The platform used here is MATLAB version 7.10, and a very elegant graphic user interface (GUI) has been developed for ease of use and aesthetic appeal. This article includes designing of a resistor value calculator and making it a standalone application using MATLAB Compiler Runtime (MCR) installer file.

Resistor colour coding

A resistor is a passive two-terminal electrical component that plays a very important role in electrical circuits. The values of most resistors is shown by colour codes or colour bands. The decoding of the resistor colour code can be done either manually or through a computer software. To manually calculate the value of resistors, we use the colour code technique shown in Fig. 1.

For example, as shown in Fig. 1, the first band is of red colour and its value is 2, the second band is violet and its value is 7, the third band (which is a multiplier) is yellow and its value is 10 raised to the power of 4. The fourth band is gold, which shows the

tolerance of resistor ($\pm 5\%$ for gold and $\pm 10\%$ for silver). This makes the

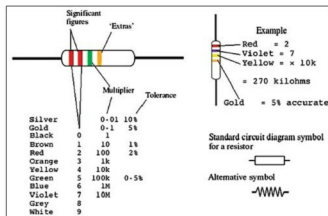


Fig. 1: Decoding resistor colour code

value of resistor $27 \times 10,000 \pm 5\% = 27k\Omega \pm 5\%$.

But what if we have hundreds of resistors in front of us and we have to find values of all of them? Then calculating the value of each resistor using the above method will be a bit difficult because for

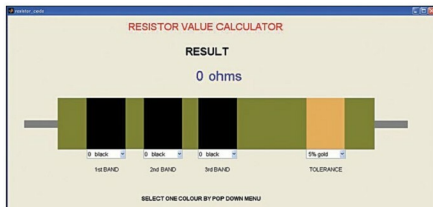


Fig. 2: Program output

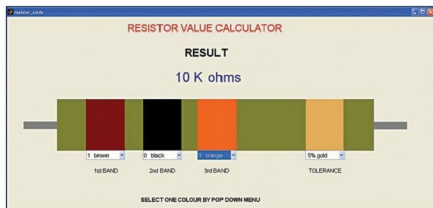


Fig. 3: Selecting the colour band for 10-kilo-ohm resistor value